

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/272167124>

UCLA 20150211

Data · February 2015

CITATIONS

0

READS

186

2 authors:



[Hyenkyun Woo](#)

Korea University of Technology and Education

30 PUBLICATIONS 309 CITATIONS

[SEE PROFILE](#)



[Haesun Park](#)

Georgia Institute of Technology

221 PUBLICATIONS 7,260 CITATIONS

[SEE PROFILE](#)

Some of the authors of this publication are also working on these related projects:



DARPA XDATA [View project](#)

Robust Asymmetric Nonnegative Matrix Factorization

Hyenkyun Woo

Korea Institute for Advanced Study

Math @UCLA
Feb. 11. 2015

Joint work with

Haesun Park Georgia Institute of Technology,

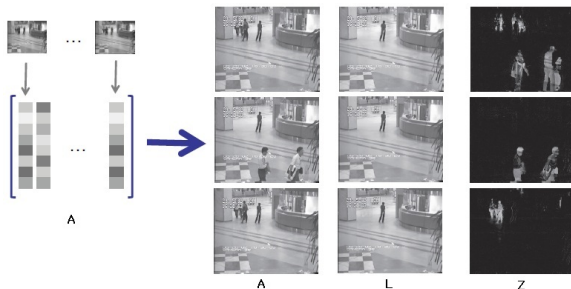
Outline

- 1 Robust Principal Component Analysis
- 2 Asymmetric Nonnegative Matrix Factorization
 - Asymmetric Nonnegative Nuclear Norm
 - Denseness and Stability
 - Asymmetric Incoherence Condition
- 3 Asymmetric Soft Regularization
- 4 Numerical Results

Outline

- 1 Robust Principal Component Analysis
- 2 Asymmetric Nonnegative Matrix Factorization
 - Asymmetric Nonnegative Nuclear Norm
 - Denseness and Stability
 - Asymmetric Incoherence Condition
- 3 Asymmetric Soft Regularization
- 4 Numerical Results

Motivation : Robust Principal Component Analysis



- Given data : A matrix ($A \geq 0$)
- Separation Problems : $A = L_0 + O_0$ (low rank) + O_0 (grouped outliers)
- Related works: Candes, Wright, Ma, Chandrasekaran, Parrilo, Tao, Recht, Osher, Yin, etc.

Goal : Asymmetric NMF for (low rank₊ + outliers)

- Background modeling by ANMF

$$\min_{L,O} \{ \|O\|_{\ell_1} + \beta \|A - L - O\|_F^2 : \text{rank}_+(L) \leq r, 0 \leq L \leq R \}$$



Figure : Proposed Model (ANMF for low rank₊ + Outliers)

- Choose relatively large $r > \text{rank}_+(L_0)$
- $\text{rank}_+(L_0) \leq r_{\text{eff}} \leq r$ is automatically recovered by Soft Regularization (or algorithmic regularization)

Review: Robust Principal Component Analysis

Robust PCA:

$$\begin{aligned} \min_L \quad & \|A - L\|_{\ell_1} \\ \text{s. t.} \quad & \text{rank}(L) \leq r. \end{aligned}$$

where $\|X\|_{\ell_1} = \sum_{ij} |X_{ij}|$.

- NP-hard problem and highly nonconvex
- How to find low rank matrix L ?
- We have relaxations: nuclear norm, γ_2 -norm, etc.

Robust PCA with nuclear norm

$$\min_L \lambda \|A - L\|_{\ell_1} + \|L\|_*$$

or

Principal component Pursuit (Candes et al. 2011):

$$\begin{aligned} (\hat{L}, \hat{O}) &= \arg \min_{L, O} \lambda \|O\|_{\ell_1} + \|L\|_* \\ \text{s. t.} \quad & O = A - L, \end{aligned}$$

where $\|L\|_* = \sum_i \sigma_i(L)$ (sum of singular values of L)

- Note that e_i is a standard basis $(0, \dots, 0, 1, 0, \dots, 0)$.
- What happen if $\hat{L} = e_i e_j^T$ and $\hat{O} = e_k e_l^T$?
- Can we fix λ for random sparse outliers $\hat{O}$?

Incoherence parameter $\mu(U)$

Definition

Let $U \subseteq \mathbb{R}^n$ with $\dim(U) = r$ and $\mathbf{P}_U$ be the orthogonal projection onto U . Then the coherence of U is defined as

$$\mu(U) = \frac{n}{r} \max_{1 \leq i \leq n} \|\mathbf{P}_U \mathbf{e}_i\|_2^2$$

where $\mathbf{e}_i$ is a standard basis.

- $\mathbf{P}_U = U(U^T U)^{-1} U^T$ (U is full rank)
- $\|\mathbf{P}_U \mathbf{e}_i\|_2^2 = \|U^T \mathbf{e}_i\|_2^2$ when $U^T U = I$
-

$$1 \leq \mu(U) \leq \frac{n}{r}$$

Incoherence condition¹

$$L_0 \in R^{n \times n} = U \Sigma V^T = \sum_{i=1}^r \sigma_i u_i v_i^T$$

where $r = \text{rank}(L_0)$ and

$$\bullet \max_i \|U^T e_i\|_2^2 \leq \frac{\mu r}{n}, \quad \max_i \|V^T e_i\|_2^2 \leq \frac{\mu r}{n}$$

and $e_i = (0, 0, \dots, 1, \dots, 0)$ is a standard coordinate basis.

Note that

$$1 \leq \mu \leq \frac{n}{r}$$

and μ should be sufficiently small.

¹Candes and Recht 09, Candes and Tao '10

When $A = L_0 + O_0$

Theorem

- L_0 : $\text{rank}(L_0) \leq \rho_r \frac{n}{\mu(\log n)^2}$
- O_0 : *random sparsity pattern* of cardinality $m_s \leq \rho_s n^2$

Then with probability $1 - O(n^{-10})$, RPCA with $\lambda = \frac{1}{\sqrt{n}}$ can separate:

$$\hat{L} = L_0, \quad \hat{O} = O_0$$

But, we do not know about incoherence condition parameter μ
($1 \leq \mu \leq \frac{n}{r}$)!!!

- How do we control μ to be **small** ?
→ In general, SVD generate very dense basis!!
- If O_0 have **grouped** outliers then how?
→ We need to tune λ^2 parameter based on size of outliers.

Goal: Column Outliers + Nonnegative Rank

Definition (Column Outliers)

Let $O \in \mathbb{R}^{m \times n}$ be grouped outliers with limited row sparsity, i.e.,

$$\max_{1 \leq i \leq m} \|\text{row}_i(O)\|_0 < \zeta n,$$

then we call O as column outliers. Here, $0 < \zeta < \frac{1}{2}$ decides sparsity level of O in row direction.

$$\min_{L, O} \{ \Phi(O) + \frac{\alpha}{2} \|A - L - O\|_F^2 : \mathcal{R}(L) \leq \tau \text{ and } 0 < L \leq B_L \} \quad (1)$$

- Φ is a sparsity enforcing function (elementwise separable, ℓ_0)
- $\ell_0 < O < \ell_{2,0}$ How to?
- Change rank constraint $\mathcal{R}(L)$ to handle column outliers O .

Outline

- 1 Robust Principal Component Analysis
- 2 Asymmetric Nonnegative Matrix Factorization**
 - Asymmetric Nonnegative Nuclear Norm
 - Denseness and Stability
 - Asymmetric Incoherence Condition
- 3 Asymmetric Soft Regularization
- 4 Numerical Results

Nonnegative Rank

- Nonnegative rank of $L \in \mathbb{R}_+^{m \times n}$:

$$\text{rank}_+(L) = \arg \min_r \{r \mid L = WH^T, W \in \mathbb{R}_+^{m \times r}, H \in \mathbb{R}_+^{n \times r}\}$$

- $0 \leq \text{rank}(L) \leq \text{rank}_+(L) \leq \min\{n, m\}$
- NP-hard Problem³
- We need a relaxation for $\text{rank}_+(L)$
- How to control denseness of L to separable ?

Review: Nuclear norm and γ_2 -norm for rank ⁵

- Nuclear norm:

$$\|L\|_* = \arg \min \left\{ \sum \lambda_i : L = \sum_i \lambda_i w_i h_i^T, \quad w_i \in B_2(m), h_i \in B_2(n) \right\}$$

$$\text{where } B_2(d) = \{x \in \mathbb{R}^d \mid \|x\|_2 = 1\}$$

- γ_2 -norm:

$$\|L\|_{\gamma_2} \approx^4 \min \left\{ \sum \lambda_i : L = \sum_i \lambda_i w_i h_i^T, \quad w_i \in B_\infty(m), h_i \in B_\infty(n) \right\}$$

$$\text{where } B_\infty(d) = \{x \in \mathbb{R}^d \mid \|x\|_\infty \leq 1\}$$

⁴ $1 \leq . \leq 2$

⁵Lee et.al. '10

Review: Nuclear norm and γ_2 -norm for rank_+

- nuclear norm ⁶:

$$\|L\|_*^+ = \arg \min \left\{ \sum \lambda_i : L = \sum_i \lambda_i w_i h_i^T, \quad w_i \in B_2^+(m), h_i \in B_2^+(n) \right\}$$

$$\text{where } B_2^+(d) = \{x \in \mathbb{R}_+^d \mid \|x\|_2 \leq 1\}$$

- γ_2 -norm:

$$\|L\|_{\gamma_2}^+ \approx \min \left\{ \sum \lambda_i : L = \sum_i \lambda_i w_i h_i^T, \quad w_i \in B_\infty^+(m), h_i \in B_\infty^+(n) \right\}$$

$$\text{where } B_\infty^+(d) = \{x \in \mathbb{R}_+^d \mid \|x\|_\infty \leq 1\}$$

Possible Candidate for rank₊ + Column Outliers

$$\min_{W \geq 0, H \geq 0, O} \|O\|_{\ell_1} + \frac{\alpha}{2} \|A - WH^T - O\|_F^2 + \gamma \Psi(WH^T)$$

- nuclear norm:

$$\Psi(WH^T) = \|WH^T\|_*^+$$

- γ_2 -norm:

$$\Psi(WH^T) = \|WH^T\|_{\gamma_2}^+$$

- But, we don't want to tune γ parameter !!
- Are W and H sufficiently **dense** for separation of **column outliers**?

Relaxation: Asymmetric Nonnegative Nuclear Norm

$$\|L\|_{\diamond} = \arg \min \left\{ \sum_i \lambda_i : L = \sum_i \lambda_i w_i h_i^T, w_i \in Z_{\eta w_i}(m), h_i \in Z_{\eta h_i}(n) \right\}$$

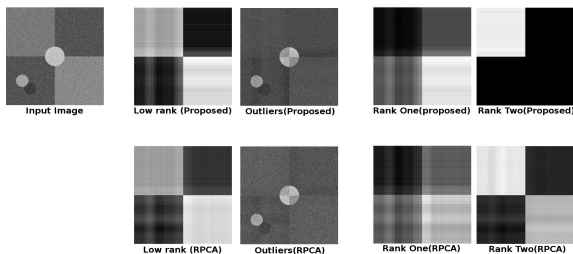
where

$$Z_{\eta}(d) = \{ v \in \mathbb{R}_+^d : v \in B_2^+(d) \cap \eta B_{\infty}^+(d) \}$$

- $B_2^+(d) = \{ v \in \mathbb{R}_+^d : \|v\|_2 = 1 \}$
- $B_{\infty}^+(d) = \{ v \in \mathbb{R}_+^d : \|v\|_{\infty} \leq 1 \}$
- $\lambda_i \geq \lambda_j$ if $i < j$
- Note that we call $L = \sum_i \lambda_i w_i h_i$ as Asymmetric NMF.
- We have many parameters : ηw_i and ηh_i .
- Why: To control denseness for separation of column outliers

Toy Model for low rank₊ + column outliers

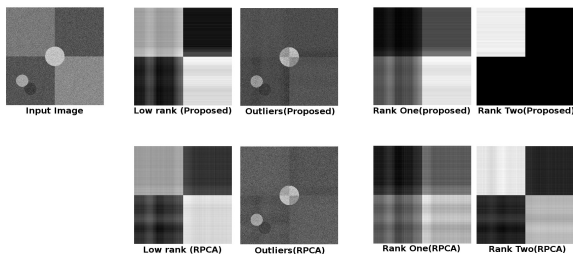
- Matrix decompositions (low rank₊ + column outliers) of the proposed Asymmetric NMF model (TOP) and Robust PCA (BOTTOM).



- Note that $A \in [0, 255]^{1000 \times 1000}$.
- Bottom: The basis of RPCA is **orthogonal** and **dense**.
- Top : The basis₊ of ANMF is **linearly independent** and **sparse**.

Toy Model for low rank₊ + column outliers

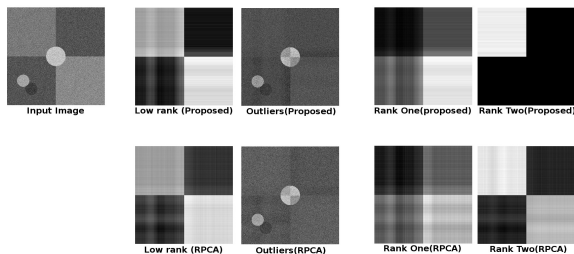
- Matrix decompositions (low rank₊ + column outliers) of the proposed Asymmetric NMF model (TOP) and Robust PCA (BOTTOM).



- Note that $A \in [0, 255]^{1000 \times 1000}$.
- Bottom: The basis of RPCA is **orthogonal** and **dense**.
- Top : The basis₊ of ANMF is **linearly independent** and **sparse**.

Toy Model for low rank₊ + column outliers

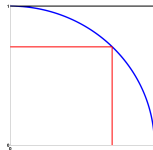
- Matrix decompositions (low rank₊ + column outliers) of the proposed Asymmetric NMF model (TOP) and Robust PCA (BOTTOM).



- Note that $A \in [0, 255]^{1000 \times 1000}$.
- Bottom: The basis of RPCA is **orthogonal** and **dense**.
- Top : The basis₊ of ANMF is **linearly independent** and **sparse**.

Why $Z_\eta(d)$: Denseness

$$\frac{1}{\sqrt{d}} B_\infty^+(d) \subseteq B_2^+(d) \subseteq 1 B_\infty^+(d)$$



where

$$B_2^+(d) = \{x \in \mathbb{R}_+^d : \|x\|_2 = 1\} \text{ and } B_\infty^+(d) = \{x \in \mathbb{R}_+^d : \|x\|_\infty \leq 1\}$$

- $v \in \frac{1}{\sqrt{d}} B_\infty^+(d) \cap B_2^+(d)$: the **unique** densest unit vector
→ dense vectors becomes unstable!!
- $v \in 1 B_\infty^+(d) \cap B_2^+(d)$: it can be standard coordinate
→ include the sparsest unit vector (i.e., $\|v\|_0 = 1$)

Why $Z_\eta(d)$: Lower bound of sparsity

Lemma

Let $w \in Z_\eta(d) = \{ v \in \mathbb{R}_+^d : v \in B_2^+(d) \cap \eta B_\infty^+(d) \}$, then

$$\|w\|_0 \geq 1/\|w\|_\infty^2 \geq 1/\eta^2$$

where $\|w\|_0 = \#\{w \neq 0\}$

Proof.

Let $\mathbf{1}_d = (1, \dots, 1)^T$ then $\forall w \in Z_\eta(d)$, we get

$$\left\langle w, \frac{1}{\sqrt{\|w\|_0}} \mathbf{1}_d \right\rangle \leq 1 \leq \langle w, \|w\|_\infty \mathbf{1}_d \rangle \leq \langle w, \eta \mathbf{1}_d \rangle$$

where $\langle w, \mathbf{1}_d \rangle > 0$.



Singular values of Asymmetric NMF ($L = \sum_i \lambda_i \mathbf{w}_i \mathbf{h}_i^T$)

Theorem

Let $L = \sum_i L_i$ with

$$\|L\|_{\diamond} = \arg \min \left\{ \sum_i \lambda_i : L_i = \lambda_i \mathbf{w}_i \mathbf{h}_i^T, \mathbf{w}_i \in Z_{\eta_{\mathbf{w}_i}}(m), \mathbf{h}_i \in Z_{\eta_{\mathbf{h}_i}}(n) \right\}.$$

Then

$$E_0(L_i) \sqrt{\|L_i\|_0} \leq \lambda_i \leq R \sqrt{\|L_i\|_0}$$

where

$$E_0(x) = \|x\|_1 / \|x\|_0 \quad \text{and} \quad R = \|L\|_{\ell_{\infty}}$$

- Approximately, we can say

$$\lambda_i \approx O(\sqrt{\|L_i\|_0})$$

Lower bound of $\text{rank}_+(L)$

Theorem

Let $L = \sum_i L_i$ have an Asymmetric NMF ($L_i = \lambda_i w_i h_i^T$) with

$$\|L\|_{\diamond} = \arg \min \left\{ \sum_i \lambda_i : w_i \in Z_{\eta_{w_i}}(m), h_i \in Z_{\eta_{h_i}}(n) \right\}.$$

Then

$$\text{rank}_+(L) \geq \frac{\|L\|_{\diamond}}{\max_i \lambda_i} \quad \text{and} \quad \text{rank}_+(L) \geq \frac{\|L\|_A}{\max_i \sqrt{\|L_i\|_{\ell_0}}}$$

where $\|L\|_A = \sum_i \sqrt{\|L_i\|_{\ell_0}}$

- **norm-based** and **combinatorial** bound for $\text{rank}_+(L)$

Stability of basis matrix

$$W \in \mathbb{Z}_{\eta_W}^{m \times r} = \{ W = [w_1, \dots, w_r] : w_i \in Z_{\eta_{W_i}}(m), \det(W^T W) \neq 0 \}$$

where $\eta_W = \max_i \eta_{W_i}$.

- Is W stable?

- $S(W) = \|W^T W - I\|_{\ell_\infty} = \max_{i \neq j} w_i^T w_j$

- $\rightarrow$ **stable** $0 \leq S(W) < 1$ **unstable**

- $\|W\| = \sqrt{\rho_{\max}(W^T W)}$

- $\rightarrow$ **stable** $1 \leq \|W\| \leq \sqrt{1 + (r-1)S(W)} < \sqrt{r}$ **unstable**

- Which one do you prefer as a measure of stability of a matrix W ?

Stability of basis matrix : $\|W\|$

$$\sqrt{\min_i c_i + \epsilon_W \left(\frac{1}{a} - 1 \right)} \leq \|W\| \leq \sqrt{\max_i c_i - \epsilon_W (1 - a)},$$

- c_i is the sum of i -th column elements of $W^T W$
- $\epsilon_W = \min_{i,j} w_i^T w_j > 0$
- When $\epsilon_W = S(W)$, we get

$$\|W\| = \sqrt{1 + (r - 1)S(W)}$$

- $\|W\|$ is more robust measure of stability of W , since it is related to all elements of W

Stability of basis matrix: δ -distinguishable

$$W \in \mathbb{Z}_{\eta_W}^{m \times r} = \{ W = [w_1, \dots, w_r] : w_i \in Z_{\eta_{w_i}}(m), \det(W^T W) \neq 0 \}$$

where $\eta_W = \max_i \eta_{w_i}$.

- Is W (or $Z_{\eta_W}(m)$) δ -distinguishable?

For any $w_i, w_j \in W$ (or $Z_{\eta_W}(m)$),

$$\|w_i - w_j\|_2 \geq \delta?$$

- Equivalently, $S(W) = \max_{i \neq j} w_i^T w_j \leq 1 - \frac{\delta^2}{2}$

Example

- For all $\delta > 0$, $\mathbb{Z}_{\frac{1}{\sqrt{m}}}^{m \times r}$ is not δ -distinguishable.

Stability vs. Denseness

Theorem

For $1 \leq k \leq m$ and $0 < \delta \leq \sqrt{\frac{2}{k}}$, we get

$$N\left(\frac{\delta}{2}, Z_{\frac{1}{\sqrt{k}}}(m)\right) \geq \binom{m}{k} \quad (2)$$

where

$$N(\varepsilon, Z_{\eta_W}(m))$$

is the *cardinality of balls* in Z_{η_W} with minimum ε -radius and no intersection each other (ε -packing).

- If we relax *denseness* (k small) then we can obtain more *stable basis* (δ big)

Example

Let $W = [w_1, \dots, w_r] \in \mathbb{Z}_{\frac{1}{\sqrt{k}}}^{m \times r}$ with $w_i \in V_k$ for all $i = 1, \dots, r$.

$$V_k = \{x = (x_1, \dots, x_m)^T \geq 0 : \|x\|_2 = 1, \|x\|_0 = k, \text{ and } x_i \in \{0, \frac{1}{\sqrt{k}}\}\}.$$

Let us assume that $\iota_{w_i}^T \iota_{w_j} = k - 1$ for all $i \neq j$. Here, ι_{w_i} is an indicator function. Then, the condition number of W becomes

$$\text{cond}(W) = \sqrt{\frac{\rho_{\max}(W^T W)}{\rho_{\min}(W^T W)}} = \sqrt{rk - r + 1} \approx O(\sqrt{rk}).$$

That is, **stability** of W depends on **rank parameter** r and **sparsity** k of each column vector w_i .

Asymmetric NMF for low rank₊ + column outliers

$$\min_{L,O} \{ \|O\|_{\ell_1} + \beta \|A - O - L\|_F^2 : L = WSH^T, W \in \mathbb{Z}_{\mathbf{1}}^{m \times r}, H \in \mathbb{Z}_{\eta_H}^{n \times r} \}$$

where

- $S = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_r)$ with $\lambda_i \geq \lambda_j$ if $i < j$.

ANMF for low rank₊ + column outliers:

- Assumption: column outliers are tall (mainly stay on column direction)
- W : **stable** (and sparse) , H : **dense** (and unstable⁷)
- Good news: $\|w_i h_i^T\|_0 \approx O(\lambda_i^2)$
- How to find a solution of the model? **Soft Regularization**

⁷Studer et.al. (2014) Democratic Representations (equal selection of basis in W)

- **Stability** of ANMF depends on stability of W (i.e., $\|W\|$ or $S(W)$):

$$E_{ij} \leq w_i^T w_j, \quad \forall i \neq j,$$

where $E_{ij} = \langle w_i h_i^T, w_j h_j^T \rangle$.

- **Denseness** of $w_i h_i^T$ depends on denseness of H (i.e., η_H):

$$\|w_i h_i^T\|_{\ell_0} = \|w_i\|_0 \|h_i\|_0 \geq \frac{1}{\eta_H^2},$$

Since $w_i \in Z_1^m$, a rank one matrix $w_i h_i^T$ can be **thin** structure in **row direction** (i.e., in h_i direction).

- **Column outliers** O need to satisfy the following condition

$$\max_i \|\text{row}_i(O)\|_0 \leq \zeta n < \frac{1}{\eta_H^2} \leq \min_i \|h_i\|_0.$$

Asymmetric Incoherence Condition

Definition

For $H \in \mathbb{Z}_{\eta_H}^{n \times r}$, let

$$\Xi(H) = \frac{\|H\|_{\ell_\infty}}{\|H\|}, \quad (3)$$

then we get $\Xi(H) \in (\frac{1}{\sqrt{rn}}, 1]$.

- It decides stability and denseness of a matrix H .
- If $\Xi(H)$ is large (≈ 1), then H is **stable** but sparse.
- If $\Xi(H)$ is small ($\approx \frac{1}{\sqrt{rn}}$), then H is **dense** but unstable.

Asymmetric Incoherence Condition

Definition

Let $L = \sum_{i=1}^r \lambda_i w_i h_i^T$ be an ANMF with basis matrix $W = [w_1, \dots, w_r]$ and coefficient matrix $H = [h_1, \dots, h_r]$. We define asymmetric incoherence criterion of L as follows:

$$\frac{1}{\sqrt{rn}} < aINC(L) = \frac{\Xi(H)}{\Xi(W)} < \sqrt{rm}$$

- It decides stability and denseness of a matrix $L = WSH^T$.
- $aINC(L) \approx \sqrt{rm}$: W dense and H stable
- $aINC(L) \approx \frac{1}{\sqrt{rm}}$: W **stable** and H **dense**

Asymmetric Incoherence Condition

Example

Let $L = WSH^T = \sum_{i=1}^r \lambda_i w_i h_i^T$ be an ideal $\frac{1}{\sqrt{k}}$ -dense vectors. That is, $W \in \mathbb{Z}_{\frac{1}{\sqrt{k_W}}}^{m \times r}$, $H \in \mathbb{Z}_{\frac{1}{\sqrt{k_H}}}^{n \times r}$, $\|w_i\|_0 = k_W$, $\|h_i\|_0 = k_H$. Let us assume that $\iota_{w_i}^T \iota_{w_j} = k_W - 1$ and $\iota_{h_i}^T \iota_{h_j} = k_H - 1$ for worst case separability. Then

$$\Xi(W) = \frac{1}{\sqrt{rk_W - r + 1}} = \frac{1}{\text{cond}(W)} \quad \text{and} \quad \Xi(H) = \dots = \frac{1}{\text{cond}(H)}.$$

and

$$aINC(L) = \frac{\text{cond}(W)}{\text{cond}(H)} = \sqrt{\frac{rk_W - r + 1}{rk_H - r + 1}} \approx \sqrt{\frac{k_W}{k_H}} = \frac{\eta_H}{\eta_W}.$$

Outline

- 1 Robust Principal Component Analysis
- 2 Asymmetric Nonnegative Matrix Factorization
 - Asymmetric Nonnegative Nuclear Norm
 - Denseness and Stability
 - Asymmetric Incoherence Condition
- 3 Asymmetric Soft Regularization
- 4 Numerical Results

ASR: Asymmetric Soft Regularization

$$\begin{cases} (\bar{W}^{k+\frac{1}{2}}, \bar{H}^{k+\frac{1}{2}}) = \underset{\bar{W} \geq 0, \bar{H} \geq 0}{\operatorname{argmin}} \{ \|\tilde{A} - \sum_{i=1}^r \bar{w}_i \bar{h}_i^T\|_F^2 : \bar{h}_i \in cB_{\infty}^+(n) \} \\ (\bar{W}^{k+1}, \bar{H}^{k+1}) = \text{BASIS}(\bar{W}^{k+\frac{1}{2}}, \bar{H}^{k+\frac{1}{2}}) \end{cases}$$

where

$$(\bar{W}, \bar{H}S) = \text{BASIS}(\bar{W}, \bar{H})$$

satisfies the following conditions:

$$\begin{cases} WSH^T = \bar{W}\bar{H}^T \\ \bar{W} = [w_1, w_2, \dots, w_r], \quad w_i \in Z_1(m) \\ \bar{H} = [h_1, h_2, \dots, h_r], \quad h_i \in Z_{\eta_{h_i}}(n) \\ \operatorname{diag}_i(S) \geq \operatorname{diag}_j(S) \quad \text{if } i < j \end{cases}$$

with an $r \times r$ diagonal matrix S .

Optimization Framework for low rank₊ + outliers

$$\begin{cases} O^{k+1} = \underset{O}{\operatorname{argmin}} \|O\|_1 + \beta \|A - O - \sum_{i=1}^r \bar{w}_i^k (\bar{h}_i^k)^T\|_F^2 \\ (\bar{W}^{k+\frac{1}{2}}, \bar{H}^{k+\frac{1}{2}}) = \underset{\bar{W} \geq 0, \bar{H}}{\operatorname{argmin}} \{ \|A - O^{k+1} - \sum_{i=1}^r \bar{w}_i \bar{h}_i^T \|_F^2 : \bar{h}_i \in cB_{\infty}^+(n) \} \\ (\bar{W}^{k+1}, \bar{H}^{k+1}) = \text{BASIS}(\bar{W}^{k+\frac{1}{2}}, \bar{H}^{k+\frac{1}{2}}) \end{cases}$$

where $(W, HS) = \text{BASIS}(\bar{W}, \bar{H})$ and $W \in \mathcal{W}_1$ and $H \in \mathcal{H}_{\eta_H}$ and

- Plug-In Outliers detector into ASR(asymmetric soft regularization)
- Now, we only need to solve L2-NMF with a box-constraint on H !!
- Asymmetric NMF : $L = WSH^T$

L2-NMF Solver: Hierarchical ALS⁸

$$\min_{W,H} \{ F(h_1, h_2, \dots, h_r; w_1, w_2, \dots, w_r) \quad : \quad w_i \in \mathbb{R}^m, \quad h_i \in \text{cB}_{\infty}^+(n) \}$$

- $F(h_1, h_2, \dots, h_r; w_1, w_2, \dots, w_r) = \|\tilde{A} - \sum_{i=1}^r w_i h_i^T\|_F^2$

- BCD Framework:

- Update H ($i=1, \dots, r$):

$$\min_{h_i \in \text{cB}_{\infty}^+(n)} F(h_1^{k+1}, h_2^{k+1}, \dots, h_i, \dots, h_r^k; w_1^k, \dots, w_r^k)$$

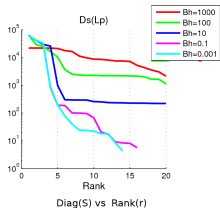
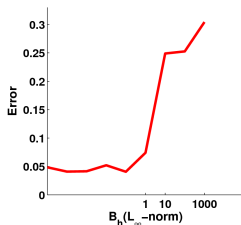
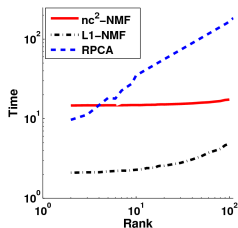
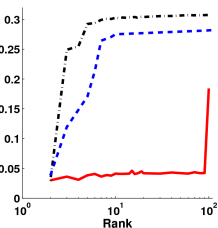
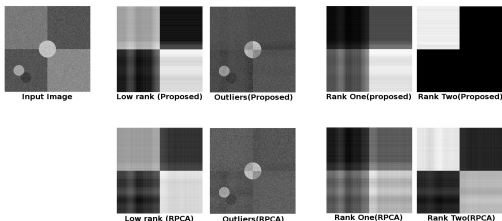
- Update W ($i=1, \dots, r$):

$$\min_{w_i \in \mathbb{R}_+^m} F(h_1^{k+1}, \dots, h_r^{k+1}; w_1^{k+1}, \dots, w_i, \dots, w_r^k)$$

Outline

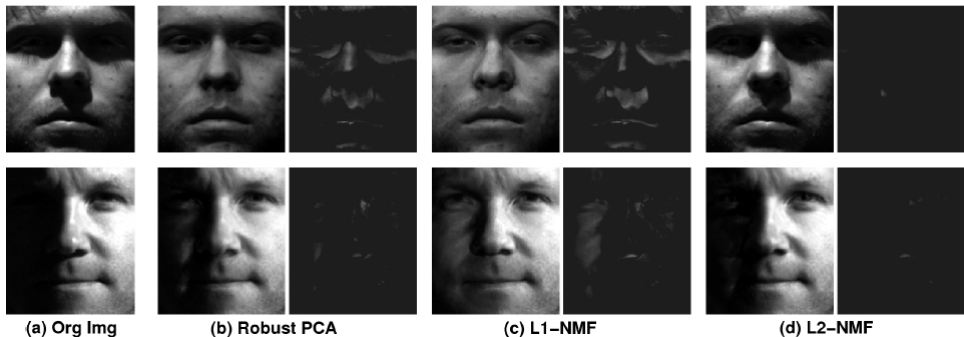
- 1 Robust Principal Component Analysis
- 2 Asymmetric Nonnegative Matrix Factorization
 - Asymmetric Nonnegative Nuclear Norm
 - Denseness and Stability
 - Asymmetric Incoherence Condition
- 3 Asymmetric Soft Regularization
- 4 Numerical Results

Numerical Results : Synthetic Image ($\text{rank}_+(L) = 2$)



- There is a strong correlation between r_{eff} and $cB_\infty^+(d)$
- small $c \rightarrow r_{\text{eff}} \gtrsim \text{rank}_+(L)$

Numerical Results ($r = 20$)

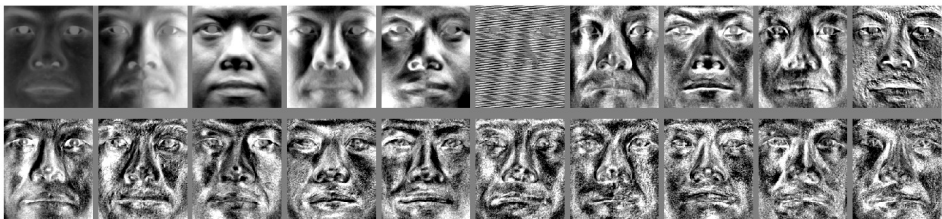


- Yale B dataset
- We have 64 different illumination direction
- A is 32256×64 matrix

20 basis of ANMF(top) vs RPCA(bottom)



Basis (Proposed)



Basis (RPCA)

20 basis of Lp-RANMF(top) vs L1-NMF(bottom)



asymmetric incoherence condition for Face Image

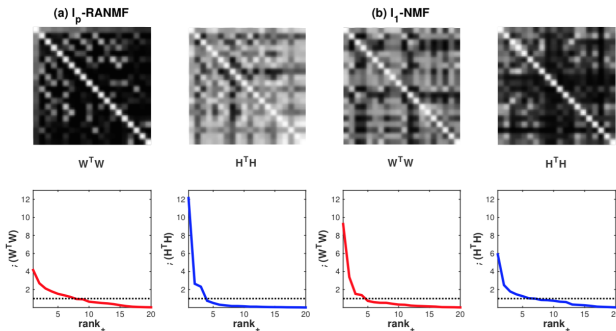
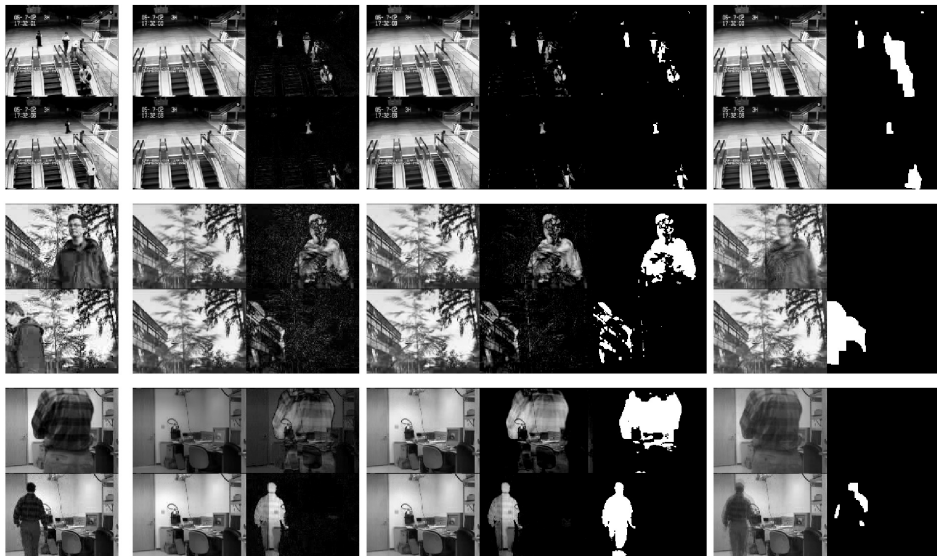


Fig. 6: A comparison of stability of (a) ℓ_p -RANMF model and (b) ℓ_1 -NMF model for face image in Figure 4 and Figure 5. It shows the gram matrix $W^T W$ and $H^T H$ and the corresponding eigenvalues $\rho(W^T W)$ and $\rho(H^T H)$. In (a) ℓ_p -RANMF, we have stable (sparse) basis matrix W and dense (unstable) coefficient matrix H ; see also Table I. In (b) ℓ_1 -NMF, as observed in Figure 5, we have dense (unstable) basis matrix W . The reason is that since $r = 20$ is relatively large compared to $n = 58$ and ℓ_1 -NMF does not have any regularization for rank_+ , each image (corresponding to column outliers) of A is included into a rank and therefore the basis matrix W of ℓ_1 -NMF is more denser and H is more sparser than that of RANMF; see also Figure 5.

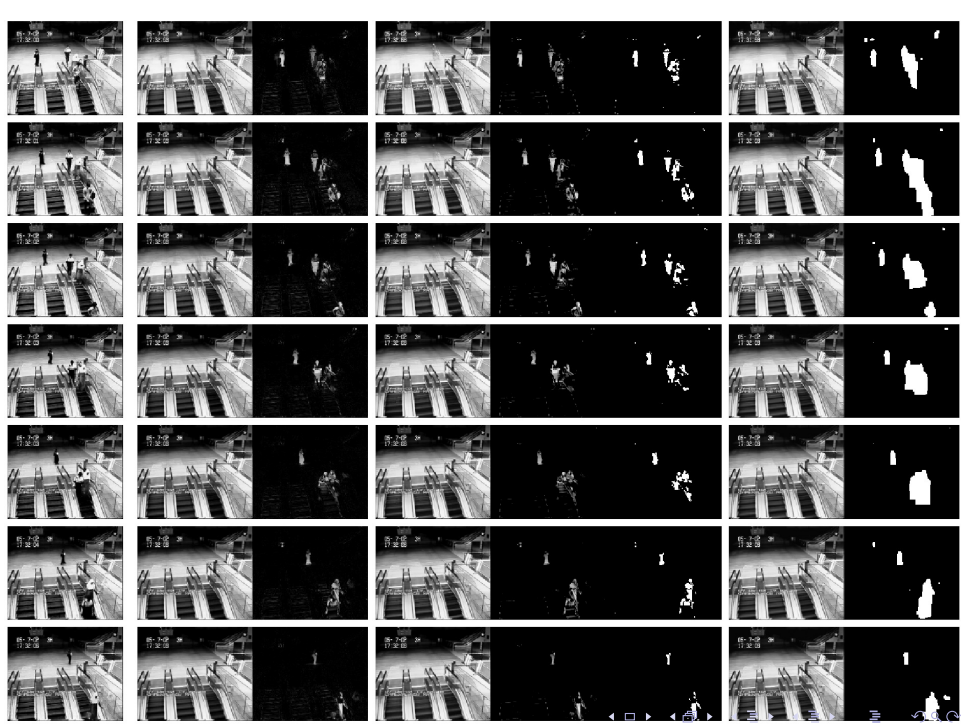
RPCA(λ is tuned for each image) vs RANMF($r = 20$)

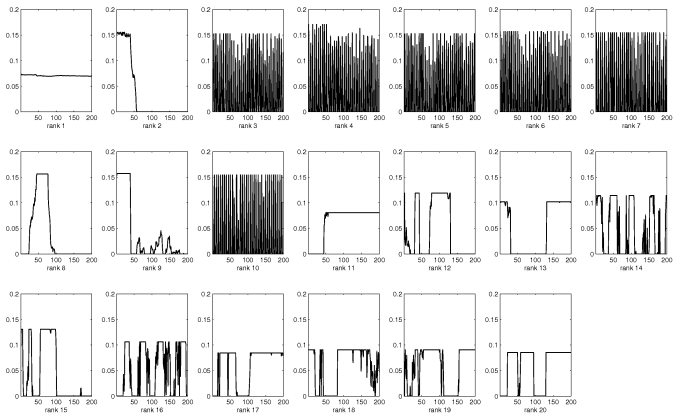
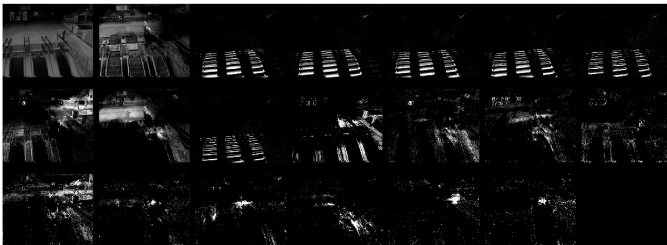


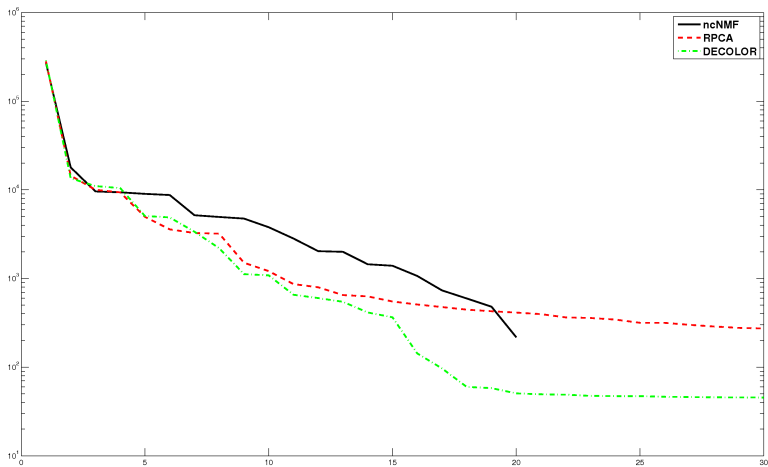
Numerical Results : $aINC(L)$

Model	$\Xi(W)$				$\Xi(H)$				$aINC(L)$			
	Face	Escalator	Tree	Office	Face	Escalator	Tree	Office	Face	Escalator	Tree	Office
ℓ_p -RANMF	0.0551	0.2596	0.1216	0.0845	0.0627	0.0505	0.0413	0.0559	1.1394	0.1944	0.3397	0.6611
$\log$ -RANMF	0.0547	0.2043	0.1348	0.3112	0.0641	0.0562	0.0417	0.0651	1.1726	0.2750	0.3095	0.2091
ℓ_1 -RANMF	0.0658	0.4004	0.2621	0.0662	0.0679	0.0581	0.0393	0.0622	1.0313	0.1450	0.1499	0.9393
ℓ_1 -NMF	0.0152	0.0348	0.0220	0.0204	0.3786	0.2291	0.2365	0.4777	24.9004	6.5751	10.7567	23.4457

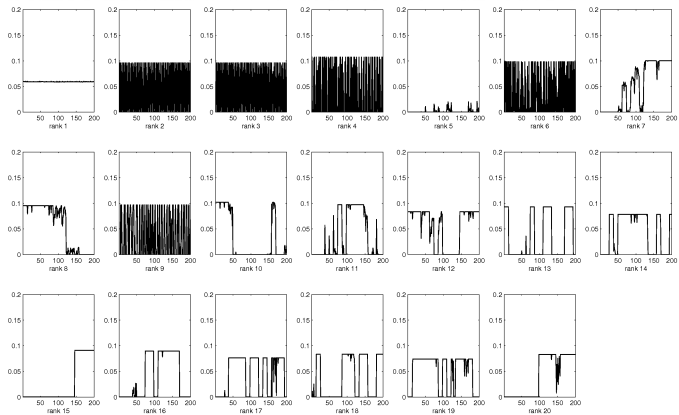
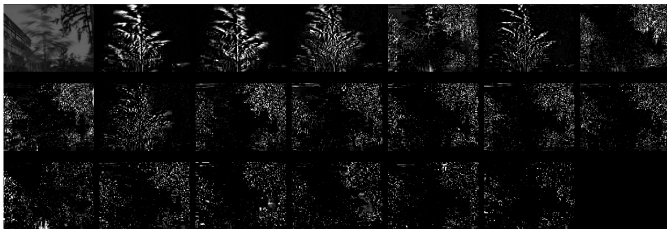
TABLE II: A comparison of $aINC(L)$ of four different models; RANMF with ℓ_p -norm ($p=0.65,1$) and $\log$ -function and ℓ_1 -NMF. Here, we use four different test image sequences; face, escalator, waving tree, and office. We can see the significant different incoherent parameter $aINC(L)$ between the proposed RANMF model and the conventional ℓ_1 -NMF model for all test image sequences. That is, $aINC(L)$ well characterize stable (and sparse) basis matrix with dense (and unstable) coefficient matrix for column outliers separation problem.

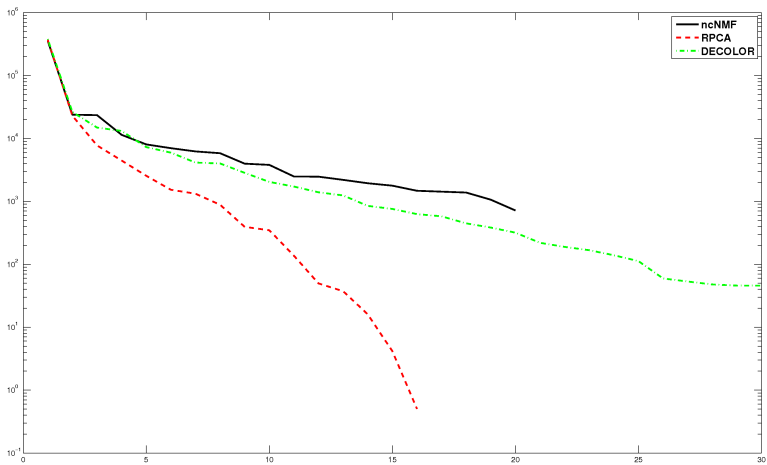




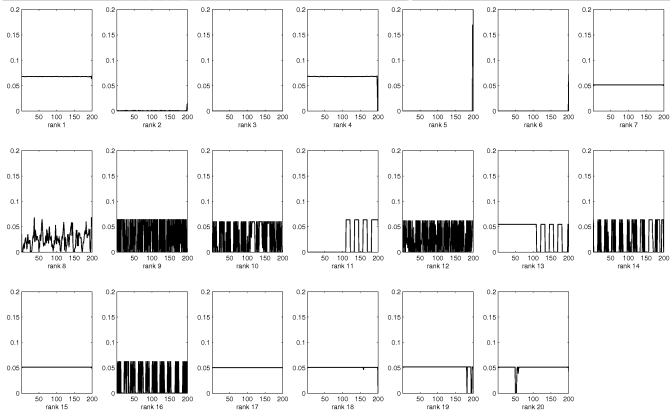
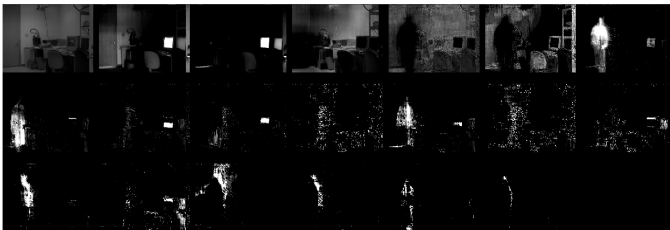


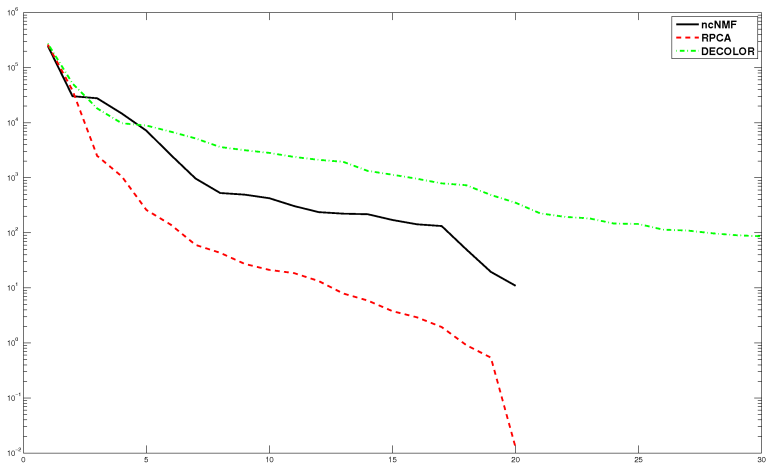












Thank you!!